

Notes: Massa (JFE, 2003)
**How Do Family Strategies Affect Fund Performance? When
Performance-Maximization Is Not the Only Game in Town**

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1 Big picture

- **Question.** How do mutual fund family strategies affect fund performance, family growth, product proliferation, and category-level competition?
- **Core idea.** Mutual fund families are not just collections of independent portfolio managers. They are multiproduct firms that choose product variety, market coverage, load-fee structures, fund entry/exit, and performance investment jointly.
- **Investor-side mechanism.** Investors differ in investment horizon, desired risk exposure, fee sensitivity, and preferred services. The ability to switch among funds inside the same family gives investors option value and gives families a way to retain assets even when one fund performs poorly.
- **Family-side mechanism.** A broad and differentiated menu can stabilize flows, reduce sensitivity to any one fund's performance, and make fund proliferation attractive. This can reduce the family's incentive to maximize every fund's risk-adjusted performance.
- **Main empirical message.** Product differentiation is negatively related to performance and positively related to fund proliferation, family category coverage, and fund turnover.
- **High-level takeaway.** Performance maximization is not the only game in town because mutual fund families compete in a differentiated-product market, not a pure return tournament.

2 Incremental contribution

- **Unit of analysis shift.** Earlier mutual fund studies mainly examine individual fund flows, expenses, and performance. This paper makes the mutual fund family the strategic unit of analysis.
- **Industrial-organization contribution.** Massa imports product differentiation, multiproduct firms, and market coverage into mutual fund performance research. The paper treats categories as markets and funds as differentiated products.
- **Alternative strategic margins.** The paper shows that a fund family can compete through product proliferation and investor retention rather than only through portfolio performance.
- **New measurement contribution.** It constructs category-level product-differentiation indexes using non-performance characteristics, performance-related characteristics, and mixed characteristics.
- **Flow-stability channel.** The paper links family breadth and load-fee structures to flow stability, showing how families internalize cross-fund externalities.
- **Dynamic panel evidence.** The empirical work uses dynamic panel GMM, lagged instruments, Sargan tests, and autocorrelation tests to study persistence and simultaneity.
- **Bridge to later literature.** The paper anticipates later research on fund-family favoritism, cross-fund subsidization, family spillovers, and fund-family agency conflicts.

Interpretation: The incremental contribution is not simply documenting that differentiated funds perform differently. The paper provides a coherent family-strategy theory and then tests its implications at fund, family-category, family-wide, category, proliferation, coverage, and turnover levels.

3 Data and measurement

3.1 Industry, categories, and families

The sample covers the U.S. mutual fund industry from 1992 to 2000. Funds are grouped into 24 categories, which the paper treats as distinct product markets. A fund family can operate in one or many categories, and each category can contain many families and many funds.

Interpretation: The category structure is central. The paper’s empirical design does not ask whether a fund beats all funds in the economy; it asks how product differentiation inside a category changes the strategic choices of families active in that category.

3.2 Fund and family variables

- **Performance:** returns or performance measures at fund, family-category, family, or category levels.
- **Risk:** standard deviation of returns over the previous 12 months.
- **Flows:** investor inflows and outflows, with volatility measured over relevant time intervals.
- **Fees:** total load fees, front-end loads, contingent deferred sales charges, redemption fees, 12b-1 fees, and expense ratios.
- **Size:** total net assets at the fund, family-category, family, or category level.
- **Proliferation:** number of funds, number of categories covered by a family, and fund turnover through births and deaths.

3.3 Product differentiation indexes

The paper uses three types of product-differentiation indexes:

- **Non-performance-related index:** dispersion in fee and service-like characteristics.
- **Performance-related index:** dispersion in performance and performance risk.
- **Mixed index:** combines performance and non-performance characteristics, and also includes distributed income, capital gain distributions, and average yields.

Each component is computed as the cross-sectional dispersion of a characteristic across funds in a category and then standardized. The paper also considers weighted and non-weighted aggregation.

Interpretation: These indexes are the paper’s key measurement innovation. They transform the abstract idea of product differentiation into data that can be used in regressions.

4 Models and empirical specifications

4.1 Flow stability

$$VF_{m,i,t} = \alpha + \beta F_{i,t} + \gamma N_{i,t} + \delta Controls_{m,i,t} + \varepsilon_{m,i,t}.$$

Here $VF_{m,i,t}$ is flow volatility, $F_{i,t}$ is a fee variable, and $N_{i,t}$ is the number of funds or family breadth. The goal is to test whether load fees and family scope stabilize investor flows.

4.2 Family-driven demand heterogeneity

The paper uses a market-share equation that is close in spirit to a nested-logit demand model:

$$\log(s_{m,i,t}) - \log(s_{0,t}) = \alpha d_{m,i,t} + \beta \log(s_{m|i,t}) + Controls + \varepsilon_{m,i,t}.$$

The equation tests whether family affiliation creates a demand nest and whether funds belonging to a common family are linked in investors’ choices.

4.3 Performance and product differentiation

$$r_{m,i,c,t} = \alpha + \beta G_{c,t} + \gamma Controls_{m,i,c,t} + \delta r_{m,i,c,t-1} + \varepsilon_{m,i,c,t}.$$

The central prediction is $\beta < 0$: in more differentiated product markets, performance competition is weaker.

4.4 Proliferation, coverage, and turnover

$$N_{c,t} = \alpha + \beta G_{c,t} + \gamma Controls_{c,t} + \delta N_{c,t-1} + \varepsilon_{c,t},$$

$$C_{i,t} = \alpha + \beta G_{i,t} + \gamma Controls_{i,t} + \delta C_{i,t-1} + \varepsilon_{i,t},$$

$$T_{c,t} = \alpha + \beta G_{c,t} + \gamma Controls_{c,t} + \delta T_{c,t-1} + \varepsilon_{c,t}.$$

The predicted sign is $\beta > 0$: differentiation supports more funds, broader family category coverage, and higher turnover.

4.5 Dynamic panel method

The paper uses dynamic panel data estimation based on GMM, White heteroskedasticity-consistent covariance matrices, and orthogonal deviations. Lagged levels and differences are used as instruments. Sargan tests and first-/second-order autocorrelation tests are reported.

5 Identification and empirical design

- **Core concern.** Performance, product dispersion, and proliferation are jointly chosen. A simple correlation cannot prove causality.
- **Design response.** The paper uses dynamic panel specifications, lagged dependent variables, lagged instruments, and several measurement definitions of product dispersion.
- **Multi-level validation.** Evidence is shown at the fund, family-category, family, category, proliferation, coverage, and turnover levels.
- **Robustness logic.** The results are more persuasive because the negative performance effect and positive proliferation effect appear across several related specifications.
- **Limit.** The design is observational. It cannot fully rule out unobserved demand shocks or category-specific technological changes, but the coherent pattern supports the family-strategy interpretation.

6 Section-by-section study guide

- **Introduction.** Motivates the puzzle of fund proliferation and explains why families may compete through product menus rather than only returns.
- **Theoretical framework.** Describes how product differentiation, investor heterogeneity, and intra-family switching change incentives.
- **Data and indexes.** Defines the 24 categories, fund and family variables, flow variables, and the three differentiation indexes.
- **Flow stability.** Establishes why family breadth and load fees matter for investor retention.
- **Performance tests.** Tables 4–8 test the negative relation between product differentiation and performance.
- **Proliferation tests.** Tables 9–11 test positive relations between differentiation and number of funds, coverage, and turnover.

7 Tables and figures

7.1 Table 1: Descriptive statistics of mutual fund industry

Read: Table 1 reports category-level counts of funds and families, performance, risk, total net assets, expense ratios, and product differentiation indexes.

Detailed interpretation: The table shows substantial heterogeneity across categories. Aggressive growth, sector, international equity, money market, bond, and utility categories differ sharply in fund counts, family counts, risk, assets, and dispersion. This heterogeneity is the raw material for the paper’s product-market tests.

Economic message: Mutual funds are differentiated products, and fund families operate in differentiated category markets.

in Panel A I report the average performance or the ratio net assets to the category and the standard deviation of their performance over time ("Risk"). I report the average value and the standard deviation in the period. In Panel C I report the value of the net total assets ("Size") of the funds belonging to the category and the average value of their expense ratios. I report the average value and the standard deviation in the period. In these three panels, I aggregate across funds by using weighted averages calculated by using as weights the total net assets values for each fund. In Panel D, I report the average value of the three types of indexes of product heterogeneity. I consider three ways of constructing the index. The first way uses only non-performance-related characteristics, that is, the dispersion of fees across all the funds in the category. The second way uses the performance-related characteristics, that is, the dispersion of fund performances and standard deviation of the performances in the previous 12 months across all the funds in the category. The third way uses both performance-related characteristics and non-performance-related ones, as defined before, along with the dispersion of the distributed incomes, capital gain distributions, and average yields. The indexes are constructed as follows. First I calculate the standard deviation of the characteristics (e.g., fees charged) across all the funds active in the category. This is done separately for each type of characteristics every year. Then, the resulting components of the index are standardized so as to make them homogeneous in terms of range of variation. The standardization is based on dividing the single components by their maximum value over the whole period. The fees I consider are maximum front-end load fees, maximum contingent deferred sales charges, rear load or redemption fees, 12b-1 fee, maximum deferred sales charges and redemption fees. I also include the expense ratio.				
Panel A				
Categories	Fund number (Average)	Fund number (Variation)	Family number (Average)	Family number (Variation)
Aggressive growth	794	179	215	41
Balanced	343	138	122	18
High quality bonds	664	205	181	22
High yield bonds	196	98	79	16
Global bonds	292	95	77	10
Global equity	305	136	79	14
Growth and income	642	253	188	18
Income	192	30	61	6
Government securities	447	88	149	10
International equities	772	424	160	41
Long-term growth	161	57	63	9
Long-term growth	1066	488	275	37
Tax-free money market	435	46	109	3
Gov securities money market	412	58	115	4
High quality municipal bond	449	111	127	7
Single-state municipal bond	1149	332	115	6
Taxable money market	494	71	171	4
High-yield money market	45	13	22	1
Precious metals	40	145	25	2
Sector	248	145	61	25
Special	14	5	3	2
Total performance	247	84	84	9
Utility	79	23	30	3

Panel B				
Categories	Performance (Average)	Performance (Variation)	Risk (Average)	Risk (Variation)
Aggressive growth	0.015	0.052	0.032	0.018
Balanced	0.009	0.022	0.024	0.007
High quality bonds	0.005	0.009	0.009	0.002
High yield bonds	0.007	0.016	0.014	0.006
Global bonds	0.004	0.012	0.015	0.004
Global equity	0.012	0.036	0.036	0.014
Growth and income	0.017	0.033	0.033	0.017
Income	0.004	0.008	0.008	0.002
Government securities	0.004	0.010	0.010	0.001
International equities	0.009	0.041	0.043	0.012
Long-term growth	0.011	0.028	0.028	0.010
Long-term growth	0.014	0.039	0.041	0.014
Tax-free money market	0.002	0.004	0.003	0.001
Gov securities money market	0.003	0.008	0.003	0.002
High quality municipal bond	0.004	0.012	0.011	0.003
Single-state municipal bond	0.004	0.012	0.012	0.004
Taxable money market	0.003	0.008	0.003	0.001
Taxable money market	0.004	0.011	0.011	0.001
Precious metals	0.002	0.091	0.088	0.031
Sector	0.018	0.050	0.057	0.022
Special	-0.001	0.023	0.022	0.012
Total performance	0.009	0.021	0.021	0.007
Utility	0.010	0.027	0.028	0.008

Panel C				
Categories	Total net assets (Average)	Total net assets (Variation)	Expense ratio (Average)	Expense Ratio (Variation)
Aggressive growth	225.78	81.43	0.012	0.003
Balanced	286.34	41.77	0.008	0.004
High quality bonds	195.21	23.30	0.007	0.003
High yield bonds	341.22	43.22	0.011	0.002
Global bonds	138.70	31.82	0.013	0.009
Global equity	311.10	95.34	0.013	0.007
Growth and income	294.48	183.91	0.007	0.004
Income	471.62	136.69	0.011	0.006
Government securities	136.31	31.18	0.009	0.006
International equities	172.75	36.57	0.013	0.007
Long-term growth	505.07	105.01	0.008	0.002
Long-term growth	399.20	136.10	0.010	0.003
Tax-free money market	313.79	58.33	0.005	0.001
Gov securities money market	541.15	99.04	0.005	0.003
High quality municipal bond	251.68	48.29	0.007	0.002
Single-state municipal bond	102.68	23.36	0.007	0.004
Taxable money market	102.11	332.35	0.005	0.001

Panel D				
Categories	Total net assets (Average)	Total net assets (Variation)	Expense Ratio (Average)	Expense Ratio (Variation)
High-yield money market	314.42	34.61	0.008	0.002
Precious metals	89.75	33.87	0.014	0.002
Sector	180.62	79.15	0.013	0.005
Special	42.85	46.11	0.012	0.009
Total performance	209.80	38.90	0.009	0.003
Utility	271.65	66.40	0.019	0.008

Panel D				
Categories	Non performance-related index		Performance-related index	
	(Average)	(Variation)	(Average)	(Variation)
Aggressive growth	0.7857	0.0362	0.3875	0.0090
Balanced	0.7940	0.0331	0.2385	0.0321
High quality bonds	0.8344	0.0315	0.2786	0.2510
High yield bonds	0.8332	0.0407	0.2339	0.0535
Global bonds	0.8125	0.0308	0.3578	0.1521
Global equity	0.8328	0.0108	0.3711	0.0741
Growth and income	0.7862	0.0643	0.2318	0.0404
Income	0.7807	0.0188	0.2198	0.0486
Government securities	0.7128	0.0343	0.2071	0.0282
International equities	0.8041	0.0317	0.3994	0.0310
Long-term growth	0.8036	0.0487	0.2745	0.0799
Tax-free money market	0.7802	0.0098	0.2670	0.0643
Gov securities money market	0.1557	0.0457	0.1996	0.0365
High quality municipal bond	0.2138	0.0853	0.1912	0.0504
Single-state municipal bond	0.7628	0.0323	0.1494	0.0393
Taxable money market	0.7848	0.0513	0.1318	0.0386
High-yield money market	0.4428	0.0843	0.2198	0.0516
Precious metals	0.0918	0.0814	0.1467	0.0318
Sector	0.0106	0.0803	0.3210	0.0987
Special	0.8034	0.0118	0.5109	0.1512
Total performance	0.1282	0.0161	0.4301	0.1453
Utility	0.2522	0.0209	0.2819	0.0814

assets in the specific month. Descriptive statistics of the categories in terms of performance, expense ratio, and number of funds and families active are provided in Table 1, Panels A, B, and C.

3.2. Construction of the index of product differentiation

I exploit the cross-sectional dimension of the mutual fund industry by considering each category as a different market. I construct an index of product dispersion (Γ) for each category, which proxies for the dispersion of the characteristics of the funds

7.2 Table 2: Stability of investors’ flows

Read: Table 2 studies flow volatility at fund, family-category, and family levels.

Detailed interpretation: The results show how load fees, family fund count, performance, risk, and family/category controls relate to investor-flow stability. The table supports the idea that families can retain investors through breadth and internal switching options.

Economic message: Stable flows are the first economic benefit of fund-family breadth.

7.4 Table 4: Fund performance and market dispersion

Read: Table 4 shows that product differentiation is negatively related to fund-level performance.

Detailed interpretation: The key coefficient on market dispersion is generally negative across weighted and non-weighted indexes and across non-performance, performance-related, and mixed index definitions. The table is the first main performance result.

Economic message: Differentiation weakens direct performance competition at the individual fund level.

Table 4
Fund performance and market dispersion
I estimate $\alpha_{n,t} = \alpha + \beta'_{n,t} + \gamma' \text{Control}_{n,t} + \delta_{n,t-1} + \epsilon_{n,t}$, where the subscripts "n", "t", "n", "t" refer, respectively, to the nth fund, of the nth family, in the nth category at time t. $\alpha_{n,t}$ is the performance of the nth fund of the nth family belonging to the nth category at time t. $\text{Control}_{n,t}$ is a vector of control variables. $\Gamma_{n,t}$ is the index of product dispersion for the nth category at time t. I consider three ways of constructing the index ("non-performance-based", "performance-based", and "mixed"). I also consider two different ways of aggregating its components: weighted average ("weighted index") and simple average ("non-weighted index"). I consider two alternative standardizations of the index of dispersion: the single components of the index are standardized using their long-term value (Panel C, "Performance and changes of market dispersion") or the average value of the components (Panels A and B, "Performance and market dispersion"). The control variables include the risk of the fund, the overall average performance of the family to which the fund belongs and of the competing families, the average risk of the family and of the competing families both overall and in the category, and the average total net assets of the family to which the fund belongs and of the competing families. The average risk is constructed as the average of the standard deviations of the 12 monthly performances. Averages are taken either across the funds operating in the same category ("in the category") or across all the funds of the same family over all categories ("overall"). The observations are yearly. The estimations are done using a Dynamic Panel Data estimation, based on GMM with instrumental variables and White's adjusted heteroskedastic consistent least-squares variance-covariance matrix. The model is estimated in Orthogonal Deviations. Lagged levels and differences of the explanatory variables are used as instruments. The Sargan test of over-identifying restrictions and tests of First Order Autocorrelation and Second Order Autocorrelation are reported. These tests are based on the standardized average residual covariances, which are asymptotically distributed $N(0,1)$ under the null of no autocorrelation. The Sargan test is asymptotically distributed as a chi-square with as many degrees of freedom as the over-identifying restrictions under the null hypothesis of the validity of the instruments. Total net assets are standardized by dividing them by 1,000,000. * means $p\text{-value} > 0.1$; ** means $p\text{-value} < 0.1$.

Panel A: Fund performance and market dispersion I

Explanatory variables	Specifications											
	Weighted Index						Non-Weighted Index					
	Non-performance-related Index		Performance-related Index		Mixed Index		Non-performance-related Index		Performance-related Index		Mixed Index	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
$\Gamma_{n,t}$	-0.08	-5.38	-0.02	-2.26	-0.02	-2.28	-0.20	-4.64	-0.01	-2.31	-0.02	-2.31
Control variables	-0.03	-0.46	-0.31	-5.71	-0.31	-5.65	-0.32	-6.54	-0.37	-7.57	-0.38	-7.57
Lagged fund performance	0.002	4.05	0.001	4.64	0.001	4.63	0.002	5.36	0.001	6.52	0.001	6.52
Family performance	-2.34	-8.83	-3.26	-6.92	-3.26	-6.94	-2.52	-11.09	-3.34	-6.91	-3.34	-6.91
Fund risk	1.17	2.17	0.09	0.18	0.11	0.21	1.61	3.77	0.31	0.81	0.31	0.81
Family risk	4.28	1.49	-2.59	-4.88	-2.59	-4.88	5.72	2.07	-2.90	-5.64	-2.90	-5.65

(a) Panel A

Average overall performance of competing families	0.67	5.13	0.46	4.32	0.47	4.35	0.33	3.65	0.38	4.58	0.38	4.58
Average risk of competing families in the category	0.93	3.25	1.97	3.40	1.96	3.37	1.84	6.74	1.94	4.53	1.94	4.53
Average overall risk of competing families	-0.24	-0.55	0.80	3.09	0.80	3.07	-1.35	-3.10	0.65	2.49	0.65	2.49
Overall total net assets for the competing families	—	—	—	—	—	—	—	—	—	—	—	—
Overall total net assets for the family	—	—	—	—	—	—	—	—	—	—	—	—
Sargan test	0.29*		11.06*		11.01*		3.09*		12.30*		12.30*	
First order autocorrelation	-0.57**		-6.54**		-6.50**		-7.02**		-6.92**		-6.92**	
Second order autocorrelation	-1.17*		0.59*		0.63*		-0.59*		0.63*		0.63*	

Panel B: Fund performance and market dispersion I

Explanatory variables	Specifications											
	Weighted Index						Non-Weighted Index					
	Non-performance-related Index		Performance-related Index		Mixed Index		Non-performance-related Index		Performance-related Index		Mixed Index	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
$\Gamma_{n,t}$	-0.74	-3.27	-0.01	-2.02	-0.02	-2.04	-0.55	-5.50	-0.02	-2.35	-0.02	-2.27
Control variables	0.86	3.00	-0.31	-4.15	-0.31	-4.15	-0.63	-4.06	-0.30	-4.07	-0.30	-4.02
Lagged fund performance	0.001	2.77	0.001	4.29	0.001	4.30	0.004	4.82	0.001	4.47	0.001	4.45
Family performance	-2.14	-1.71	-3.29	-6.39	-3.29	-6.40	-1.97	-3.24	-3.22	-6.62	-3.24	-6.66
Fund risk	8.28	1.07	-0.09	-0.30	-0.08	-0.26	6.19	2.36	0.23	0.64	0.16	0.31
Family risk	2.32	0.80	-2.44	-4.88	-2.44	-4.90	5.65	1.44	-2.38	-4.68	-2.39	-4.64
Fund total load fees	3.99	5.92	0.48	3.76	0.48	3.77	-0.18	-0.71	0.51	4.05	0.51	4.03
Average overall performance of competing families	-0.77	-0.84	2.25	5.05	2.24	5.03	-0.59	-1.81	1.76	4.08	1.84	2.75
Average risk of competing families in the category	-3.74	-0.60	0.79	3.04	0.79	3.03	-4.62	-2.13	0.84	3.13	0.85	3.12
Average overall risk of competing families	-0.04	-0.93	0.22	0.40	0.11	0.40	-0.01	-1.21	-0.11	-0.62	-0.12	-0.42
Overall total net assets for the competing families	3.24	0.11	-12.23	-1.41	-12.34	-1.41	105	3.62	-14.13	-1.57	-15.10	-1.60
Overall total net assets for the family	—	—	—	—	—	—	—	—	—	—	—	—
Sargan test	0.27*		10.81*		10.75*		0.17*		8.67*		8.32*	

(b) Panel B

Table 4 (continued)

First order autocorrelation	-1.94**		-0.51**		-6.53**		-4.47**		-6.96**		-6.97**	
Second order autocorrelation	0.02*		0.30*		0.32*		-0.15*		0.89*		-0.56*	

Panel C: Fund performance and market dispersion II

Explanatory variables	Specifications											
	Weighted Index						Non-Weighted Index					
	Non-performance-related Index		Performance-related Index		Mixed Index		Non-performance-related Index		Performance-related Index		Mixed Index	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
$\Gamma_{n,t}$	-0.88	-2.09	-0.01	-2.07	-0.02	-2.08	-0.48	-4.79	-0.02	-2.36	-0.02	-2.38
Control variables	1.13	1.99	-0.30	-4.12	-0.30	-4.06	-0.78	-3.95	-0.30	-4.07	-0.30	-4.11
Lagged fund performance	0.001	1.42	0.001	4.32	0.001	4.30	0.005	4.84	0.001	4.47	0.001	4.49
Family performance	-3.45	-1.48	-3.27	-6.44	-3.26	-6.37	-3.21	-3.21	-3.22	-6.63	-3.24	-6.62
Fund risk	14.38	0.99	-0.07	-0.23	-0.09	-0.28	7.77	2.30	0.24	0.65	0.22	0.62
Fund total load fees	2.83	0.67	-2.43	-4.89	-2.40	-4.63	6.19	1.41	-2.38	-4.68	-2.38	-4.69
Average overall performance of competing families	3.95	3.70	0.49	3.86	0.49	3.88	-0.55	-1.70	0.51	4.05	0.50	4.02
Average risk of competing families in the category	-0.68	-0.51	2.22	5.04	2.21	4.99	-0.69	-2.57	1.77	4.08	1.80	4.17
Average overall risk of competing families overall	-8.66	-0.75	0.79	3.06	0.81	2.97	-6.10	-2.16	0.83	3.12	0.82	3.05
Overall total net assets for the competing families	0.2	0.17	0.1	0.33	0.3	0.32	-0.1	-0.18	-0.1	-0.60	-0.1	-0.49
Overall total net assets for the family	-14.10	-0.36	-12.45	-1.42	-12.10	-1.28	127.12	3.79	-14.78	-1.57	-15.34	-1.59
Sargan test	0.06*		10.85*		10.69*		0.02*		8.64*		8.45*	
First order autocorrelation	-1.31**		-6.59**		-6.55**		-4.06**		-6.97**		-6.94**	
Second order autocorrelation	0.79*		0.29*		0.27*		-0.06*		0.90*		0.83*	

Table 4, Panel C.

7.5 Table 5: Family performance in the category and market dispersion

Read: Table 5 repeats the performance analysis at the family-category level.

Detailed interpretation: The results show that the negative relation between dispersion and performance survives aggregation from individual funds to family activity within a category.

Economic message: Family category performance is shaped by category product differentiation.

Table 5

Family performance in the category and market dispersion

I estimate $r_{i,t} = \alpha + \beta r_{i,t} + \gamma \text{Control}_{i,t} + \delta r_{i,t} + \epsilon_{i,t}$, where the subscripts "i," "t," and "i" refer, respectively, to the i-th family, in the i-th category at time t. $r_{i,t}$ is the average performance of all the funds the i-th family has in the i-th category at time t. $\text{Control}_{i,t}$ is a vector of control variables. $r_{i,t}$ is the index of product dispersion for the i-th category at time t. I consider three ways of constructing the index ("non-performance-based," "performance-based," and "mixed"). I also consider two different ways of aggregating its components, weighted average ("weighted index") and simple average ("non-weighted index"), and two alternative standardizations of the index of dispersion: the single components of the index are standardized using their long-term value ("Performance and changes of market dispersion") or the average value of the components ("Performance and market dispersion"). The control variables include the risk, and expense ratio of the family in the category, the overall family performance, risk, and expense ratio, the performance, risk, and expense ratio of the competing families in the category, and the overall performance, risk and expense ratios across categories. I also consider the total load fees charged by the family as well as those charged by the competing families. The average Risk is constructed as the average of the standard deviations of the 12 monthly performances. Averages are taken either across the funds operating in the same category ("in the category") or across all the funds of the same family over all categories ("overall"). The observations are yearly. The estimations are done using a Dynamic Panel Data estimation, based on GMM with instrumental variables and White's adjusted heteroskedastic consistent least squares variance-covariance matrix. The model is estimated in Orthogonal Deviations. Lagged levels, and differences of the explanatory variables are used as instruments. The Sargan test of over-identifying restrictions and tests of First Order Autocorrelation and Second Order Autocorrelation are reported. These tests are based on the standardized average residual covariances, which are asymptotically distributed $N(0,1)$ under the null of no autocorrelation. The Sargan test is asymptotically distributed as a chi-square with as many degrees of freedom as the over-identifying restrictions under the null hypothesis of the validity of the instruments. * means p-value > 0.1; ** means p-value < 0.1.

Panel A: Family performance in the category and market dispersion I

Explanatory variables	Specifications											
	Weighted Index				Non-Weighted Index							
	Performance-related Index		Mixed Index		Performance-related Index		Performance-related Index		Mixed Index		Mixed Index	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
$r_{i,t}$	-0.04	-2.38	-0.005	-2.13	-0.006	-2.61	-0.14	-2.47	-0.02	-2.32	-0.02	-2.32
Lagged family performance in the category	0.13	1.98	-0.01	-0.53	-0.02	-1.06	0.03	0.62	0.08	1.12	0.08	1.11
Overall family performance	0.81	6.07	0.27	1.43	0.31	1.79	0.80	6.17	0.79	4.55	0.79	4.56
Average performance of competing families in the category	0.17	0.58	0.93	24.62	0.91	23.26	0.57	0.83	0.23	0.39	0.23	0.40
Overall average performance of competing families	0.18	0.50	-0.21	-1.29	-0.24	-1.55	-0.12	-0.39	0.06	0.14	0.06	0.14

(a) Panel A, part 1

Family risk in the category	-0.76	-0.97	-0.62	-2.51	-0.67	-2.83	-1.05	-2.02	-1.19	-1.65	-1.19	-1.65
Overall family risk	0.69	1.06	-0.23	-1.21	-0.26	-1.72	1.05	2.63	1.01	1.94	1.01	1.95
Average risk of competing families in the category	0.30	1.54	0.65	2.54	0.69	2.94	0.24	1.36	0.31	1.33	0.31	1.33
Overall average risk of competing families	-0.24	-1.47	0.21	1.28	0.26	1.81	-0.43	-1.73	-0.12	-0.39	-0.13	-0.40
Total load fees of the family in the category	0.58	0.87	0.05	0.54	0.08	0.85	1.08	1.80	1.04	1.41	1.05	1.43
Total load fees of the competing families in the category	-1.81	-0.84	-0.10	-0.29	-0.23	-0.57	-5.94	-2.70	-1.99	-1.06	-2.02	-1.08
Expense ratio of the family in the category	0.03	0.02	-0.73	-0.70	-0.63	-0.77	0.60	0.49	0.76	0.52	0.77	0.53
Overall expense ratio of the family	-0.31	-0.46	0.72	0.57	0.51	0.46	-0.57	-0.72	-0.65	-0.64	-0.66	-0.87
Expense ratio of competing families in the category	0.53	0.06	1.44	1.47	0.97	1.14	-1.68	-0.29	6.59	0.73	6.56	0.73
Overall expense ratio of competing families	-0.47	-0.09	-0.92	-0.89	-0.49	-0.51	-2.13	-0.48	-4.84	-0.83	-4.82	-0.83
Sargan test	2.08*		5.48*		4.08*		4.18*		2.04*		2.06*	
First order autocorrelation	-2.05**		-4.54**		-5.02**		-3.02**		-1.91**		-1.92**	
Second order autocorrelation	0.39*		0.71*		0.81*		0.81*		0.87*		0.88*	

Panel B: Family performance in the category and market dispersion II

$r_{i,t}$	-0.04	-2.29	-0.005	-2.28	-0.006	-2.64	-0.10	-2.92	-0.01	-2.33	-0.01	-2.35
Lagged family performance in the category	0.12	1.97	-0.01	-0.59	-0.01	-1.02	0.05	0.98	0.08	1.10	0.08	1.10
Overall family performance	0.81	6.21	0.24	1.38	0.31	1.82	0.72	6.69	0.78	4.55	0.78	4.60
Average performance of competing families in the category	0.20	0.48	0.94	24.92	0.91	21.40	0.75	2.46	0.22	0.39	0.23	0.40

(b) Panel A, part 2

Table 5 (continued)

Explanatory variables	Specifications											
	Weighted Index				Non-Weighted Index							
	Non-performance-related Index		Performance-related Index		Mixed Index		Non-performance-related Index		Performance-related Index		Mixed Index	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Average performance of competing families overall	0.12	0.35	-0.21	-1.36	-0.24	-1.61	-0.35	-1.75	0.06	0.13	0.05	0.12
Family risk in the category	-0.79	-1.02	-0.50	-2.39	-0.67	-2.84	-0.66	-1.95	-1.19	-1.65	-1.19	-1.66
Overall family risk	0.72	1.12	-0.09	-0.58	0.26	-1.71	0.81	2.67	1.01	1.94	1.01	1.96
Average risk of competing families in the category	0.29	1.52	0.51	2.38	0.69	2.95	0.12	0.88	0.31	1.33	0.31	1.34
Average risk of competing families overall	-0.24	-1.51	0.08	0.64	0.25	1.79	-0.46	-2.25	-0.12	-0.39	-0.13	-0.42
Total load fees of the family in the category	0.65	0.99	0.01	0.12	0.08	0.82	1.20	2.24	1.04	1.41	1.05	1.44
Total load fees of the competing families in the category	-1.58	-0.74	0.13	0.42	-0.22	-0.56	-4.78	-2.78	-1.98	-1.06	-2.01	-1.08
Expense ratio of the family in the category	0.08	0.06	-0.19	-0.19	-0.56	-0.68	0.44	0.39	0.76	0.53	0.80	0.56
Expense ratio of competing families in the category	-0.34	-0.50	0.16	0.13	0.37	0.34	-0.70	-0.88	-0.65	-0.87	-0.67	-0.90
Expense ratio of competing families overall	0.11	0.01	0.86	0.96	0.83	0.97	-8.33	-1.46	6.57	0.73	6.45	0.72
Overall expense ratio of competing families	-0.04	-0.01	-0.30	-0.32	-0.28	-0.30	2.09	0.68	-4.79	-0.82	-4.71	-0.82
Sargan test	2.55*		8.28*		4.45*		4.69*		2.07*		2.15*	
First order autocorrelation	-2.07**		-4.95**		-5.03**		-4.48**		-1.91**		-1.94**	
Second order autocorrelation	0.18*		0.40*		0.76*		0.22*		0.87*		0.88*	

Table 5, Panel B.

7.6 Table 6: Family overall performance and market dispersion

Read: Table 6 moves to overall family performance.

Detailed interpretation: Product differentiation remains negatively related to performance when the family is treated as the unit of analysis. This supports the claim that family-wide strategies, not only fund-specific choices, affect returns.

Economic message: Differentiated family product menus can lower overall family performance.

Table 6
Family overall performance and market dispersion

I estimate $r_{i,t} = \alpha + \beta'_{i,t} + \gamma \text{Controls}_{i,t} + \delta_{i,t} + \epsilon_{i,t}$, where the subscripts "i" and "t" refer, respectively, to the *i*th family at time *t*. $r_{i,t}$ is the average performance of the all the funds that the *i*th family manages across all the categories at time *t*. **Controls**_{*i,t*} is a vector of control variables. $I_{i,t}$ is the index that proxies for the average degree of product dispersion of all the categories the family has funds in at time *t*. I consider three ways of constructing the index ("non-performance-based," "performance-based," and "mixed"), and two different ways of aggregating its components: weighted average ("weighted index") and simple average ("non-weighted index"). I also consider two alternative standardizations of the index of dispersion: the single components of the index are standardized using their long-term value ("Performance and changes of market dispersion") or using the average value of the components ("Performance and market dispersion"). For each family the family-specific index of product dispersion is constructed by weighting the indexes of all the categories in which the family is operating, by the total net assets of the funds it is running in such categories. The control variables include the average performance of the family and of the competing families and the average total load fees and expense ratios of both the family and the competing families. The observations are yearly. The estimations are done using a Dynamic Panel Data estimation, based on GMM with instrumental variables and White's adjusted heteroskedastic consistent least squares variance-covariance matrix. The model is estimated in Orthogonal Deviations. Lagged levels and differences of the explanatory variables are used as instruments. The Sargan test of over-identifying restrictions and tests of First Order Autocorrelation and Second Order Autocorrelation are reported. These tests are based on the standardized average residual covariances, which are asymptotically distributed N(0,1) under the null of no autocorrelation. The Sargan test is asymptotically distributed as a chi-square with as many degrees of freedom as the over-identifying restrictions under the null hypothesis of the validity of the instruments. * means *p*-value > 0.1; ** means *p*-value < 0.1.

Panel A: Family performance and market dispersion I

Explanatory variables	Specifications											
	Weighted Index						Non-Weighted Index					
	Non-performance-related Index			Performance-related Index			Non-performance-related Index			Performance-related Index		
	Value	t-stat		Value	t-stat		Value	t-stat		Value	t-stat	
$I_{i,t}$	-0.31	-2.73	-0.02	-2.21	-0.19	-2.70	-0.66	-2.56	-0.03	-4.62	-0.02	2.99
Control variables	-0.35	-2.33	-0.05	-0.97	-0.73	-2.94	-1.28	-3.17	-0.31	-4.59	0.38	1.46
Lagged family performance	-5.94	-1.58	4.43	5.38	16.14	2.51	-3.72	-2.71	-4.87	-4.39	-8.63	-4.29
Total load fees of competing families	-1.02	-0.74	0.02	0.01	2.02	1.40	2.58	0.60	0.25	0.32	-0.76	-1.77
Expense ratio of competing families	34.77	2.72	3.49	1.56	20.24	2.79	12.54	2.49	5.41	4.14	5.36	4.36
Sargan test	0.06*			0.78*			0.13*			0.09*		
First order autocorrelation	-4.35**			-3.09**			-2.49**			-1.79**		
Second order autocorrelation	-0.05*			-0.42*			-0.33*			1.07*		

(a) Panel A

Table 6 (continued)

Panel B: Family performance and market dispersion II

Explanatory variables	Specifications											
	Weighted Index						Non-Weighted Index					
	Non-performance-related Index			Performance-related Index			Non-performance-related Index			Performance-related Index		
	Value	t-stat		Value	t-stat		Value	t-stat		Value	t-stat	
$I_{i,t}$	-0.47	-2.67	-0.03	-3.23	-0.35	-2.18	-0.46	-2.78	-0.03	-4.61	-0.02	-3.01
Control variables	-0.22	-1.21	-0.08	-1.42	-1.03	-2.21	-1.20	-3.53	-0.31	-4.59	0.37	1.44
Lagged family performance	2.34	3.11	1.15	12.67	1.38	4.66	0.91	5.03	0.97	22.67	1.69	6.21
Average performance of competing families	-3.23	-2.17	-0.01	-0.10	0.06	0.13	-2.40	-1.41	-0.27	-0.96	-0.17	-0.49
Family total load fees	-10.86	-1.90	3.94	4.02	27.94	2.13	-30.36	-2.98	-4.90	-4.39	-8.77	-4.31
Total load fees of competing families	0.18	0.10	-0.20	-0.13	3.06	1.35	2.20	0.55	0.26	0.33	-0.76	-1.77
Expense ratio of competing families	57.54	2.65	5.56	2.28	55.21	2.19	12.83	2.86	5.51	4.15	5.96	4.36
Sargan test	0.02*			0.50*			0.22*			0.09*		
First order autocorrelation	-3.92**			-2.87**			-2.38**			-1.65**		
Second order autocorrelation	-2.32*			-0.36*			-0.90*			-1.02*		

(b) Panel B

7.7 Table 7: Category performance and market dispersion

Read: Table 7 estimates category-level performance regressions.

Detailed interpretation: The negative relation between category performance and dispersion indicates that the performance consequences of differentiation are visible even at the market/category level.

Economic message: More differentiated categories tend to offer lower average performance.

Table 7
Category performance and market dispersion

I estimate $r_{i,t} = \alpha + \beta'_{i,t} + \gamma \text{Controls}_{i,t} + \delta_{i,t} + \epsilon_{i,t}$, where the subscripts "i" and "t" refer, respectively, to the *i*th category at time *t*. $r_{i,t}$ is the average performance of all funds in the *i*th category at time *t*. **Controls**_{*i,t*} is a vector of control variables below. $I_{i,t}$ is the index that proxies for the average degree of product dispersion of the *i*th category at time *t*. I consider three ways of constructing the index ("non-performance-based," "performance-based," and "mixed"), and two different ways of aggregating its components: weighted average ("weighted index") and simple average ("non-weighted index"). I also consider two alternative standardizations of the index of dispersion: the single components of the index are standardized using their long-term value ("Performance and changes of market dispersion") or the average value of the components ("Performance and market dispersion"). The control variables include the three Fama and French factors (i.e. Market, SMB and HML) and the riskless rate (i.e., performance on the three-month T-Bill rate). I also include the average risk of the category. The average Risk is constructed as the average of the standard deviations of the 12 monthly performance of all the funds in the category. Averages are taken across all the funds in the category. The observations are monthly. The estimations are done using a Dynamic Panel Data estimation, based on GMM with instrumental variables and White's adjusted heteroskedastic consistent least squares variance-covariance matrix. The model is estimated in Orthogonal Deviations. Lagged levels and differences of the explanatory variables are used as instruments. The Sargan test of over-identifying restrictions and tests of First Order Autocorrelation and Second Order Autocorrelation are reported. These tests are based on the standardized average residual covariances, which are asymptotically distributed N(0,1) under the null of no autocorrelation. The Sargan test is asymptotically distributed as a chi-square with as many degrees of freedom as the over-identifying restrictions under the null hypothesis of the validity of the instruments. * means *p*-value > 0.1; ** means *p*-value < 0.1.

Panel A: Category performance and market dispersion I

Explanatory variables	Specifications											
	Weighted Index						Non-Weighted Index					
	Non-performance-related Index			Performance-related Index			Non-performance-related Index			Performance-related Index		
	Value	t-stat		Value	t-stat		Value	t-stat		Value	t-stat	
$I_{i,t}$	-0.10	-3.13	-3.75	-7.54	-3.47	-8.25	-0.10	-8.39	-6.77	-4.54	-7.14	-6.41
Control variables	-0.26	-6.92	-0.40	-7.74	-0.43	-7.32	-0.28	-14.22	-0.39	-8.10	-0.43	-9.79
Lagged performance	3.92	4.29	4.29	5.40	4.64	5.60	4.09	8.27	5.21	8.81	6.15	8.06
Risk	0.002	2.36	0.004	10.66	0.004	10.08	0.003	6.75	0.006	9.70	0.006	8.70
Market	-0.0004	1.14	0.001	4.74	0.002	6.00	0.001	6.78	0.002	7.13	0.003	8.19
SMB	-0.0007	-1.07	-0.006	-1.45	-0.002	-0.55	0.001	3.04	0.006	0.93	0.007	1.21
HML	0.10	6.38	0.14	3.37	0.17	4.65	0.10	2.18	0.27	8.25	0.28	7.90
Riskless rate	10.98*		9.65*		10.32*		17.44*		12.29*		12.32*	
Sargan test	-2.51**			-2.99**			-3.02**			-3.44**		
First order autocorrelation	0.84*			-0.32*			-0.09*			0.70*		
Second order autocorrelation												

(a) Panel A

Table 7 (continued)

Panel B: Category performance and market dispersion II

Explanatory Variables	Specifications											
	Weighted Index						Non-Weighted Index					
	Non-performance-related Index			Performance-related Index			Non-performance-related Index			Performance-related Index		
	Value	t-stat		Value	t-stat		Value	t-stat		Value	t-stat	
$I_{i,t}$	-0.10	-3.28	-0.58	-2.86	-0.32	-2.56	-0.10	-8.22	-1.25	-3.86	-1.20	-5.29
Control variables	-0.26	-6.95	-0.29	-6.94	-0.27	-8.66	-0.28	-14.06	-0.34	-3.21	-0.34	-3.38
Lagged performance	3.85	4.19	2.73	3.14	2.31	2.98	4.06	8.19	3.67	4.48	3.71	4.75
Market	0.002	2.39	0.003	4.58	0.002	3.05	0.003	6.91	0.007	6.29	0.007	6.85
Risk	0.0005	1.39	-0.00003	-0.74	0.0008	2.71	0.001	6.31	0.0007	1.13	0.0008	1.26
SMB	-0.0007	-1.19	-0.0006	-0.89	-0.001	-3.04	0.001	2.80	0.002	3.52	0.003	3.74
HML	0.10	6.83	0.14	6.49	0.008	5.11	0.10	6.93	0.18	7.57	0.20	8.04
Riskless Rate	11.29*		11.29*		10.17*		17.53*		14.35*		15.12*	
Sargan test	-2.51**			-2.41**			-2.40**			-3.22**		
First order autocorrelation	0.84*			0.609*			0.67*			0.73*		
Second order autocorrelation												

(b) Panel B

7.8 Table 8: Causality between performance and market dispersion

Read: Table 8 estimates a dynamic system linking performance and dispersion.

Detailed interpretation: The causality tests help address whether poor performance creates differentiation or differentiation drives performance. The results provide evidence that dispersion affects performance dynamics.

Economic message: The negative performance-dispersion relation is not only a static correlation.

Table 8
Causality between performance and market dispersion

I estimate the following system of equations:

$$r_{i,t} = \sum_{k=1}^K \beta_{k,i,t-1} + \sum_{k=1}^K \beta_{k,i,t-2} + \eta_{i,t} + \epsilon_{i,t}$$

$$\Gamma_{i,t} = \sum_{k=1}^K \beta_{k,i,t-1} + \sum_{k=1}^K \beta_{k,i,t-2} + \eta_{i,t} + \epsilon_{i,t}$$

where $r_{i,t}$ is the average performance of the funds belonging to the i th category at time t , $\Gamma_{i,t}$ is the degree of product dispersion of i th category at time t , $\eta_{i,t}$ represents a category specific effect, and $\epsilon_{i,t}$ is the number of lags. I consider three ways of constructing the index ("non-performance-based", "performance-based", and "mixed") and two different ways of aggregating its components: weighted average ("weighted index") and simple average ("non-weighted index"). I also consider two alternative standardizations of the index of dispersion: the single components of the index are standardized using their long-term value ("Performance and changes of market dispersion") or the average value of the components ("Performance and market dispersion"). The control variables include the past performance of the category, the three Fama and French factors (i.e. Market, SMB, and HML) and the riskless rate (i.e. return on the three-month T-bill rate). I also include the average risk of the category. The average Risk is constructed as the average of the standard deviations of the 12 monthly performances of all the funds in the category. Averages are taken across all the funds in the category. The observations are yearly. The estimations are done using a Dynamic Panel Data estimation, based on GMM with instrumental variables and White's adjusted heteroskedastic consistent least squares variance-covariance matrix. The model is estimated in Orthogonal Deviations. Lagged levels and differences of the explanatory variables are used as instruments. The Sargan test of over-identifying restrictions and tests of First Order Autocorrelation and Second Order Autocorrelation are reported. These tests are based on the standardized average residual covariances which are asymptotically distributed $N(0,1)$ under the null of no autocorrelation. The Sargan test is asymptotically distributed as a chi-square with as many degrees of freedom as the over-identifying restrictions under the null hypothesis of the validity of the instruments. I report the χ^2 and the p -values of the test of causality. * means p -value > 0.1; ** means p -value < 0.1.

Panel A: Causality between performance and market dispersion I

	Weighted index	Non-Weighted index										
	Non-performance-related index	Performance-related index	Non-performance-related index	Performance-related index	Mixed index							
	χ^2	p-values	χ^2	p-values	χ^2	p-values	χ^2	p-values	χ^2	p-values		
Dispersion > performance	36.09	0.00	19.44	0.00	20.32	0.00	12.38	0.00	4.31	0.22	9.69	0.02
Sargan	3.80		2.22		1.80		6.76		6.09		4.44	
Test of first order autocorrelation	-2.84		-1.95		-2.15		-1.82		-5.45		-5.98	

(a) Panel A

Test of second order autocorrelation	-0.75	-0.81	-0.71	0.47	-0.82	0.38						
Performance = > dispersion	5.50	0.13	40.39	0.00	40.15	0.00	5.71	0.12	100.29	0.00	85.21	0.00
Sargan	4.45*	1.74*	1.93*	4.15*	3.52*	3.08*						
Test of first order autocorrelation	-1.75**	-4.23**	-4.28**	-3.49**	-2.23**	-2.63**						
Test of second order autocorrelation	0.19*	1.25*	1.18*	0.31*	0.56*	0.62*						

Panel B: Causality between performance and market dispersion II

Dispersion = > performance	29.61	0.00	27.56	0.00	27.36	0.00	15.67	0.00	8.07	0.04	8.45	0.03
Sargan	2.59		2.88		3.55		8.90		2.51		2.49	
Test of first order autocorrelation	-2.08		-1.66		-1.66		-1.71		-6.22		-6.13	
Test of second order autocorrelation	-0.94		-0.65		-0.89		-0.71		-1.32		-1.25	
Performance = > dispersion	2.72	0.43	24.65	0.00	20.45	0.00	7.77	0.09	39.01	0.00	23.10	0.00
Sargan	1.69*	0.85*	0.90*	3.53*	10.61*	0.81*						
Test of first order autocorrelation	-1.65**	-2.64**	-2.59**	-2.26**	-1.74**	-1.78**						
Test of second order autocorrelation	0.42*	0.04*	0.003*	1.01*	-0.12*	-0.50*						

(b) Panel B

7.9 Table 9: Fund proliferation: Number of funds and market dispersion

Read: Table 9 shows that categories with higher dispersion tend to have more funds.

Detailed interpretation: This is the first product-proliferation result. The positive coefficients on dispersion imply that product differentiation supports more active fund creation.

Economic message: Product variety is an active strategic margin for mutual fund families.

Table 9
Fund proliferation: Number of funds and market dispersion
I estimate $N_{i,t} = \alpha + \beta \Gamma_{i,t} + \gamma \text{Control}_{i,t} + \delta N_{i,t-1} + \epsilon_{i,t}$, where the subscripts "i" and "t" refer, respectively, to the i th family at time t . $N_{i,t}$ is the number of funds of the i th family in the i th category at time t . $\text{Control}_{i,t}$ is a vector of control variables. $\Gamma_{i,t}$ is the index of the product dispersion of the i th category at time t . I consider three ways of constructing the index ("non-performance-based", "performance-based", and "mixed") and two different ways of aggregating its components: weighted average ("weighted index") and simple average ("non-weighted index"). I also consider two alternative standardizations of the index of dispersion: the single components of the index are standardized using their long-term value ("Performance and changes of market dispersion") or the average value of the components ("Performance and market dispersion"). The control variables include the average performance of the family and of the competing families in the category, the average total load fees and expense ratios of both the family and the competing families in the category and the family overall performance. Averages are taken across all the funds of the same family over all categories ("overall"). The observations are yearly. The estimations are done using a Dynamic Panel Data estimation, based on GMM with instrumental variables and White's adjusted heteroskedastic consistent least squares variance-covariance matrix. The model is estimated in Orthogonal Deviations. Lagged levels and differences of the explanatory variables are used as instruments. The Sargan test of over-identifying restrictions and tests of First Order Autocorrelation and Second Order Autocorrelation are reported. These tests are based on the standardized average residual covariances which are asymptotically distributed $N(0,1)$ under the null of no autocorrelation. The Sargan test is asymptotically distributed as a chi-square with as many degrees of freedom as the over-identifying restrictions under the null hypothesis of the validity of the instruments. * means p -value > 0.1; ** means p -value < 0.1.

Panel A: Number of funds and market dispersion I

Explanatory variables	Specifications											
	Weighted index						Non-Weighted index					
	Non-performance-related index			Performance-related index			Non-performance-related index			Performance-related index		
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
$\Gamma_{i,t}$	5.69	2.15	4.50	3.84	5.33	4.11	67.98	2.33	3.88	3.89	4.01	3.89
Control variables												
Lagged number of funds	0.25	1.98	0.08	1.13	-0.003	-0.03	1.10	4.42	0.06	0.86	0.06	0.86
Family performance in the category	-0.87	-1.97	-0.81	-2.22	-0.77	-2.01	1.71	1.61	-1.78	-4.84	1.78	-4.84
Overall family performance	-0.35	-0.02	-36.49	-3.17	-46.46	-3.62	-138.51	-1.92	-40.50	-3.35	-41.04	-3.37
Average performance of competing families in the category	29.27	0.17	331.80	3.24	361.69	3.36	119.86	0.29	345.61	3.31	345.36	3.31

(a) Panel A

Family total load fees	-397.84	-1.64	-1171.35	-5.57	-1259.00	5.77	2301.69	1.87	-1286.07	-5.79	-1279.27	-5.78
Total load fees of competing families	50.64	0.69	36.55	0.90	38.77	0.87	787.94	0.98	32.73	0.82	32.81	0.82
Family expense ratio in the category	-48.25	-0.10	751.33	2.05	797.23	2.07	-913.35	-0.98	1394.36	4.45	1391.82	4.44
Expense ratio of competing families in the category	0.0002	1.63	0.0001	1.78	0.0001	1.83	0.0001	0.68	0.0001	1.52	0.0001	1.52
Sargan test	4.47*		11.98*		6.41*		7.23*		10.71*		10.73*	
First order autocorrelation	-4.22**		-5.19**		-1.66**		-5.02**		-5.27**		-5.27**	
Second order autocorrelation	0.90*		0.71*		0.15*		1.49*		0.57*		0.57*	

Panel B: Number of funds and market dispersion II

$\Gamma_{i,t}$	5.71	2.11	4.37	3.86	5.13	4.13	55.90	2.37	3.93	3.89	4.07	3.89
Control variables												
Lagged number of funds	0.23	1.81	0.08	1.10	-0.002	-0.03	1.11	4.41	0.06	0.85	0.06	0.85
Family performance in the category	-0.82	1.88	-0.94	-2.66	-0.93	-2.54	1.86	1.69	-1.77	-4.83	-1.77	-4.83
Overall family performance	-0.74	-0.05	-35.32	-3.11	-44.48	-3.55	-152.77	-2.01	-40.35	-3.34	-41.05	-3.38
Average performance of competing families in the category	30.14	0.18	330.80	3.24	357.28	3.34	139.85	0.33	344.96	3.31	344.68	3.31
Family total load fees	-425.23	-1.80	-1167.96	-5.57	-1248.41	-5.76	2540.81	1.93	-1280.67	-5.79	-1271.69	-5.77
Family total load fees of competing families	51.21	0.70	35.80	0.88	38.07	0.86	789.05	0.99	32.78	0.82	32.88	0.82
Family expense ratio in the category	-50.51	-0.10	746.42	2.04	795.01	2.07	-1189.47	-1.18	1385.95	4.42	1381.69	4.41
Expense ratio of competing families in the category	0.0002	1.67	0.0001	1.79	0.0001	1.84	0.0001	0.65	0.0001	1.53	0.0001	1.53
Sargan test	4.47*		11.98*		6.54*		7.23*		10.73*		10.73*	
First order autocorrelation	-4.21**		-5.19**		-1.66**		-5.02**		-5.27**		-5.27**	
Second order autocorrelation	0.79*		0.69*		0.15*		1.54*		0.57*		0.56*	

(b) Panel B

7.10 Table 10: Fund proliferation: Family coverage and market dispersion

Read: Table 10 studies how product dispersion affects the number of categories a family covers.

Detailed interpretation: Family coverage is positively related to family-level market dispersion. This supports the idea that families expand breadth when differentiation creates investor-retention value.

Economic message: Family scope is part of the product-differentiation strategy.

Table 10
Fund Proliferation: Family coverage and market dispersion

I estimate $C_{it} = \alpha + \beta'U_{it} + \gamma \text{Controls}_{it} + \delta C_{it-1} + e_{it}$, where the subscripts "i" and "t" refer, respectively, to the i th family at time t . C_{it} is the degree of market coverage, the number of categories in which the i th family is present at time t . Controls_{it} is a vector of control variables. F_{it} is the index that proxies for the average degree of product dispersion of all the categories in which the family has funds at time t . I consider three ways of constructing the index ("non-performance based," "performance-based," and "mixed") and two different ways of aggregating its components: weighted average ("weighted index") and simple average ("non-weighted index"). I also consider two alternative standardizations of the index of dispersion: the single components of the index are standardized using their long-term value ("Performance and changes of market dispersion") or the average value of the components ("Performance and market dispersion"). The family specific index of product dispersion is constructed by weighting the indices of all the categories in which the family is operating by the total net assets of the funds it is running in such categories. The control variables include the average overall performance of the family and of the competing families and the average total load fees and expense ratios of both the family and the competing families. Averages are taken across all the funds of the same family over all categories ("overall"). The observations are yearly. The estimations are done using a Dynamic Panel Data estimation, based on GMM with instrumental variables and White's adjusted heteroskedastic consistent least squares variance-covariance matrix. The model is estimated in Orthogonal Deviations. Lagged levels and differences of the explanatory variables are used as instruments. The Sargan test of over-identifying restrictions and tests of First Order Autocorrelation and Second Order Autocorrelation are reported. These tests are based on the standardized average residual covariances, which are asymptotically distributed $N(0,1)$ under the null of no autocorrelation. The Sargan test is asymptotically distributed as a chi-square with as many degrees of freedom as the over-identifying restrictions under the null hypothesis of the validity of the instruments. * means $p\text{-value} < 0.1$, ** means $p\text{-value} < 0.01$.

Panel A: Family coverage and market dispersion I

Explanatory variables	Specifications					
	Weighted index			Non-Weighted index		
	Non-performance-related index	Performance-related index	Mixed index	Non-performance-related index	Performance-related index	Mixed index
	Value	t-stat	Value	t-stat	Value	t-stat
F_{it}	0.17	2.78	0.08	2.54	0.08	2.55
Control variables						
Lagged family coverage	0.44	1.96	-0.01	-0.5	-0.01	-0.08
Family performance	1.38	4.46	-0.16	-0.77	-0.15	-0.73
Average performance of competing families	1.79	-3.97	-0.18	-0.72	-0.21	-0.80
Family total load fees	0.15	0.47	-4.82	-3.43	-4.77	-3.46

(a) Panel A

Total load fees of competing families	4.24	1.83	-4.03	-1.36	-3.94	-1.35	4.29	2.78	-1.96	-1.92	-1.94	-1.87
Family expense ratio	1.62	1.96	1.36	1.09	1.39	1.13	0.43	0.20	2.83	1.57	2.85	1.52
Expense ratio of competing families	-17.30	-2.67	-9.71	-2.14	-9.99	-2.17	-2.51	-1.58	-0.65	-0.45	0.62	0.43
Sargan test	0.53*		7.97*		8.18*		6.28*		0.67*		0.80*	
First order autocorrelation	-6.05**		-4.15**		-4.19**		-5.04**		-1.81**		-1.67**	
Second order autocorrelation	1.15*		0.05*		0.02*		1.11*		1.00*		0.97*	

Panel B: Family coverage and market dispersion II

F_{it}	0.11	2.55	0.09	2.82	0.09	2.82	0.0005	4.95	0.0003	4.89	0.0003	4.79
Control variables												
Lagged family coverage	0.48	2.68	0.09	0.49	0.07	0.41	0.37	1.50	0.36	3.20	0.34	3.14
Family performance	1.37	3.93	-0.15	-0.77	-0.13	-0.70	-0.03	-0.19	0.38	1.35	0.39	1.34
Average performance of competing families	-1.56	-3.53	-0.24	-1.01	-0.28	-1.11	0.08	0.65	-0.47	-1.50	-0.49	-1.52
Family total load fees	0.08	0.27	-4.37	-3.64	-4.32	-3.66	-2.75	-5.20	0.16	0.24	0.15	0.23
Total load fees of competing families	1.01	0.76	-5.73	-1.80	-5.48	-1.77	4.45	2.91	-2.01	-1.91	-1.95	-1.83
Family expense ratio	0.95	0.61	1.19	1.05	1.24	1.11	0.34	0.16	2.94	1.57	2.91	1.48
Expense ratio of competing families	-11.43	-2.04	-12.50	-2.49	-12.94	-2.51	-2.67	-1.67	-0.63	-0.42	-0.61	-0.40
Sargan test	1.29*		8.28*		8.56*		6.25*		0.75*		0.96*	
First order autocorrelation	-7.13**		-4.47**		-4.51**		-4.76**		-1.72**		-1.57**	
Second order autocorrelation	1.54*		0.39*		0.25*		1.10*		1.00*		0.87*	

(b) Panel B

7.11 Table 11: Fund turnover and market dispersion

Read: Table 11 studies births and deaths of funds by category.

Detailed interpretation: Fund turnover is positively related to product dispersion. Differentiated categories encourage product experimentation because investors can distinguish products and families can replace unsuccessful products with lower reputational spillovers.

Economic message: Differentiation raises the option value of launching and closing funds.

Table 11
Fund turnover and market dispersion

I estimate $T_{it} = \alpha + \beta'U_{it} + \gamma \text{Controls}_{it} + \delta T_{it-1} + e_{it}$, where the subscripts "i" and "t" refer, respectively, to the i th category at time t . T_{it} is the number of new and dead funds in the i th category at time t . Controls_{it} is a vector of control variables. F_{it} is the index that proxies for the average degree of product dispersion of the i th category at time t . I consider three ways of constructing the index ("non-performance based," "performance-based," and "mixed") and two different ways of aggregating its components: weighted average ("weighted index") and simple average ("non-weighted index"). I also consider two alternative standardizations of the index of dispersion: the single components of the index are standardized using their long-term value ("Performance and changes of market dispersion") or the average value of the components ("Performance and market dispersion"). The control variables include the overall performance of the mutual fund industry, the excess return on the market portfolio, the average risk of the category, and the lagged values of the number of fund in the category and of the total net assets of all the funds operating in the category. The average Risk is constructed as the average of the standard deviations of the 12 monthly returns of all the funds in the category. Averages are taken across all the funds in the category. The observations are monthly. The estimations are done using a Dynamic Panel Data estimation, based on GMM with instrumental variables and White's adjusted heteroskedastic consistent least squares variance-covariance matrix. The model is estimated in Orthogonal Deviations. Lagged levels and differences of the explanatory variables are used as instruments. The Sargan test of over-identifying restrictions and tests of First Order Autocorrelation and Second Order Autocorrelation are reported. These tests are based on the standardized average residual covariances which are asymptotically distributed $N(0,1)$ under the null of no autocorrelation. The Sargan test is asymptotically distributed as a chi-square with as many degrees of freedom as the over-identifying restrictions under the null hypothesis of the validity of the instruments. * means $p\text{-value} > 0.1$, ** means $p\text{-value} < 0.01$.

Panel A: Fund turnover and market dispersion I

Explanatory variables	Specifications					
	Weighted index			Non-Weighted index		
	Non-performance-related index	Performance-related index	Mixed index	Non-performance-related index	Performance-related index	Mixed index
	Value	t-stat	Value	t-stat	Value	t-stat
F_{it}	47.48	3.47	692.48	3.87	708.35	4.04
Control variables						
Lagged turnover	-0.27	-3.52	-0.99	-4.98	-0.29	-5.16
Performance	1.07	0.18	11.27	0.38	3.20	0.08
Risk	126.52	0.92	-263.50	-1.62	-260.36	-1.60
Overall performance	117.58	3.13	113.65	3.09	127.68	3.24
Market	-0.38	-3.21	-0.59	-4.63	-0.59	-4.65
Lagged total assets	-0.002	-2.95	-0.001	-0.66	-0.001	-0.81

(a) Panel A

Lagged number of funds	0.008	1.11	0.02	1.38	0.02	1.36	0.001	2.46	0.11	2.66	0.09	2.55
Sargan test	15.23*		11.88*		12.43*		13.31*		6.10*		7.03*	
First order autocorrelation	-3.50**		-3.39**		-3.38**		-3.39**		-3.67**		-3.84**	
Second order autocorrelation	-0.63*		-0.69*		-0.77*		-1.57*		0.87*		0.71*	

Panel B: Fund turnover and market dispersion II

F_{it}	43.57	3.71	137.17	2.55	133.54	3.21	9.37	3.43	409.04	2.34	479.07	3.06
Control variables												
Lagged turnover	-0.27	-3.58	-0.01	0.13	-0.06	-0.71	-0.36	-5.33	0.11	1.05	0.006	0.64
Performance	5.08	0.29	-12.48	-1.47	-12.53	-1.46	-1.20	-0.08	-15.83	-1.74	-15.92	-1.68
Risk	117.79	0.85	-405.85	-1.85	-389.89	-1.86	179.37	1.23	-103.19	-1.22	-103.95	-1.15
Overall performance	117.08	3.30	-171.30	-0.59	-170.99	-0.60	112.46	3.41	-136.16	-0.71	-149.85	-0.77
Market	-0.37	-3.13	1.11	0.84	1.11	0.85	-0.33	-2.79	0.95	1.10	1.04	1.17
Lagged total assets	-0.002	-2.81	-0.0001	-0.04	-0.0001	-0.08	0.002	-3.43	-0.001	-1.20	-0.0007	-0.02
Lagged number of funds	0.008	1.12	0.002	0.35	-0.006	-0.70	0.01	2.30	0.0003	0.02	-0.009	-0.74
Sargan test	15.00*		6.37*		4.74*		13.47*		5.85*		5.38*	
First order autocorrelation	-3.55**		-3.26**		-3.18**		-3.32**		-3.32**		-3.43**	
Second order autocorrelation	-0.57*		1.38*		1.20*		-1.44*		1.57*		1.45*	

(b) Panel B

8 Result map

- **Flow stability channel:** families use breadth and load-fee structures to stabilize investor flows.
- **Demand nesting channel:** family affiliation matters in investor choice, so funds inside a family are strategic complements.
- **Performance channel:** product differentiation lowers performance at fund, family-category, family, and category levels.

- **Causality channel:** dynamic system tests support dispersion affecting performance, not merely reflecting performance.
- **Proliferation channel:** dispersion raises the number of funds, family coverage, and fund turnover.
- **Overall strategic lesson:** families trade off performance enhancement against product-menu breadth and investor-retention services.

9 Short summary

- Massa studies mutual fund families as strategic multiproduct firms.
- Product differentiation is measured using non-performance, performance-related, and mixed dispersion indexes.
- Differentiation helps stabilize flows and creates demand segmentation.
- Differentiation is negatively related to performance.
- Differentiation is positively related to fund proliferation, family category coverage, and turnover.
- The central lesson is that mutual fund families may maximize family-level value rather than the performance of every individual fund.

10 Conclusion

10.1 Core conclusion

Massa shows that mutual fund family strategy affects performance and industry structure. Families do not compete only by maximizing returns; they compete through differentiated product menus, internal switching options, market coverage, and fund proliferation.

10.2 Economic intuition

Investors do not buy raw performance alone. They buy bundles of services, categories, fee structures, and switching options. A family that offers many differentiated products can retain assets even when a particular fund underperforms. This lowers the marginal benefit of investing in performance enhancement and raises the benefit of product proliferation.

10.3 Incremental contribution revisited

The paper's incremental contribution is to connect mutual fund performance research with industrial organization. The family becomes the strategic firm, categories become product markets, and product differentiation becomes a measurable determinant of performance, proliferation, coverage, and turnover.

10.4 Implications

For investors, family strategy may matter as much as individual fund performance. For researchers, fund-level performance tests should account for family-level product-market strategy. For regulators, proliferation and menu complexity may affect investor welfare even when every fund is individually disclosed.

10.5 Limitations

The analysis is observational and relies on product-differentiation proxies. It cannot fully eliminate unobserved demand shocks or category-specific technological changes. The welfare effect is also ambiguous because product variety provides choice value even if it weakens performance incentives.

10.6 Takeaway

Performance maximization is not the only game in town. Mutual fund families are multiproduct intermediaries, and their product-differentiation strategies shape flows, performance, and the structure of the industry.